

PROJECT DESIGN REPORT (PDR)

Project Title: Automated Sentiment Intelligence Engine (ASIE)

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Category: MLOps / AI Infrastructure / Distributed Systems

1. EXECUTIVE SUMMARY

The Automated Sentiment Intelligence Engine (ASIE) is a **production-oriented MLOps system** designed to address one of the most persistent challenges in applied machine learning: **model degradation under real-world data shift**.

While traditional DevOps practices ensure code stability, machine learning systems require additional guarantees around **data integrity, model validity, and continuous adaptation**. ASIE is a **Level 2 MLOps system**, that demonstrates how these guarantees can be achieved through a layered, incremental architecture that evolves from reproducible training to a fully automated, closed-loop deployment system.

Rather than optimizing for model accuracy alone, ASIE prioritizes:

- reproducibility
- observability
- operational safety
- controlled automation

The project is intentionally scoped as a **12-week system build**, reflecting how real-world ML platforms are developed over time rather than in a single leap.

2. PROBLEM STATEMENT & OBJECTIVE

2.1 The Problem

Machine learning models degrade as the data they operate on changes. This phenomenon – often referred to as **model decay** – is especially severe in domains with rapidly evolving language, such as a financial sentiment analysis.

For example, a model trained on expert-labeled financial news may perform well initially, yet fail silently when exposed to:

- Informal investor commentary
- Emerging slangs
- New financial narratives

Standard DevOps monitoring (CPU, memory, uptime) remains green, while **business value quietly collapses**.

2.2 Objective

ASIE aims to address this gap by building a system that

- **Establishes strong foundations** for reproducible training and data versioning
- **Serves models safely and observably** in production environments.
- **Detects data and prediction drifts** before performance collapses
- **Automates retraining and redeployment** using cloud-native infrastructure
- **Balances automation with control**, allowing human intervention wherever necessary

3. DATA ARCHITECTURE

3.1 Baseline Dataset (Training)

- **Source:** Financial PhraseBank (sentences_allagree).
- **Volume:** ~2,300 expert-labeled financial news sentences.
- **Canonical Format:** Parquet
- **Versioning:** DVC (Data Version Control)
- **Storage:** Object storage (local during development, S3 in cloud)

The dataset is treated as an immutable, versioned artifact. All preprocessing, splits, and transformations are derived from this canonical dataset.

3.2 Inference Data (Live Inputs)

- **Source:** External financial text streams (**NewsAPI**).
- **Usage:** Real-time inference via API
- **Ingestion:** Python-based ingestion engine running as a CronJob on **AWS EKS**.
- **Persistence:** Sampled inference data stored for monitoring and analysis.

Inference data is never used directly for training without passing through validation and versioning stages

3.3 Synthetic Drift (Validation)

To validate the monitoring and retraining pipeline, ASIE introduces **synthetic distribution shifts**, such as:

- Informal language
- Social media-style slang
- Vocabulary mismatches

This allows controlled testing of drift detection and retraining logic without waiting for real-world failures.

4. TECHNICAL SPECIFICATIONS & TOOLS

<u>Layer</u>	<u>Technology</u>	<u>Justification</u>
Data Versioning	DVC	Strong dataset identity and reproducibility
Experiment Tracking	MLflow	Model lineage, metrics, and registry
Model Serving	FastAPI	Explicit, low-latency inference API
Containerization	Docker	Environment parity and portability
Orchestration	Kubernetes (EKS)	Scalable, production-grade runtime
Workflow Orchestration	Apache Airflow	Event-driven retraining pipelines
GPU Optimization	NVIDIA CUDA/cuDNN	Packaged via Docker to ensure environment parity between local and cloud.
Monitoring	Evidently AI	Specialized in detecting Kolmogorov-Smirnov (KS) data drift and target drift.
Deployment	Helm + GitOps	Declarative, auditable deployments

5. SYSTEM ARCHITECTURE (THE CLOSED LOOP)

1. **Prediction:** The FastAPI service on EKS serves the DistilBERT model.
2. **Observation:** Each request is sampled and sent to a "Shadow Bucket" in S3.
3. **Analysis:** An **Evidently AI** service compares the "Shadow Bucket" against the "DVC Baseline."
4. **Trigger:** If the drift score exceeds 0.5, a webhook fires to **Apache Airflow**.
5. **Execution:** Airflow provisions a transient **EC2 G4dn** instance, fetches the new data, retrains the model, and pushes the new version to **MLflow**.
6. **Deployment:** **ArgoCD** detects the model update and performs a rolling update of the EKS pods.

6. IMPLEMENTATION ROADMAP (12 WEEKS)

Weeks 1-2: Foundations

Week 1 – Reproducible Training Core

Week 2 – Data Versioning and Registry

Weeks 3-4: Serving

Week 3 – Inference Service

Week 4 – Containerization and Shadow Traffic

Weeks 5-6: Infrastructure

Week 5 – AWS Foundations and IAM

Week 6 – Kubernetes (EKS) and Helm

Weeks 7-8: Observability

Week 7 – Drift Detection

Week 8 – Alerts and Triggers

Weeks 9-10: Orchestration

Week 9 – Automated Retraining

Week 10 – GPU and Cost Optimization

Weeks 11-12: Deployment and Hardening

Week 11 – GitOps Deployment

Week 12 – Closed Loop and Demo

7. RISK MITIGATION & SECURITY

- **Data poisoning:** human-in-the-loop model promotion
- **Cost overruns:** spot instances and controlled GPU usage
- **Automation failures:** explicit gates and rollback mechanisms
- **Over-complexity:** strict week-by-week scope discipline

8. CONCLUSION

ASIE is not a single tool or a pipeline – it is a **system engineering exercise** in how machine learning should be built, deployed, and evolved in production.

The system emphasizes correctness before automation, observability before scale, and clarity over cleverness.